GSAS @NIDA : Deep Learning

MLP 1

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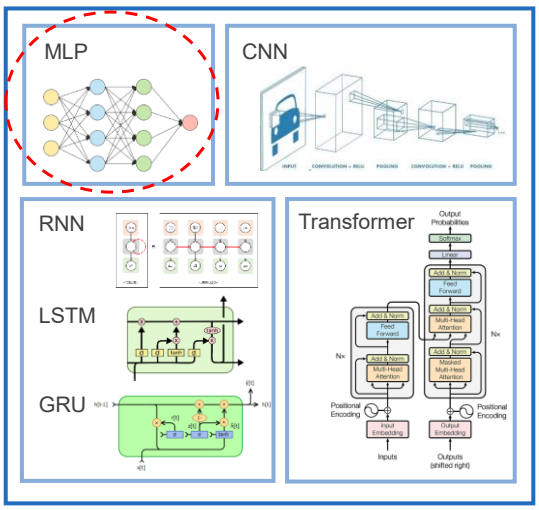
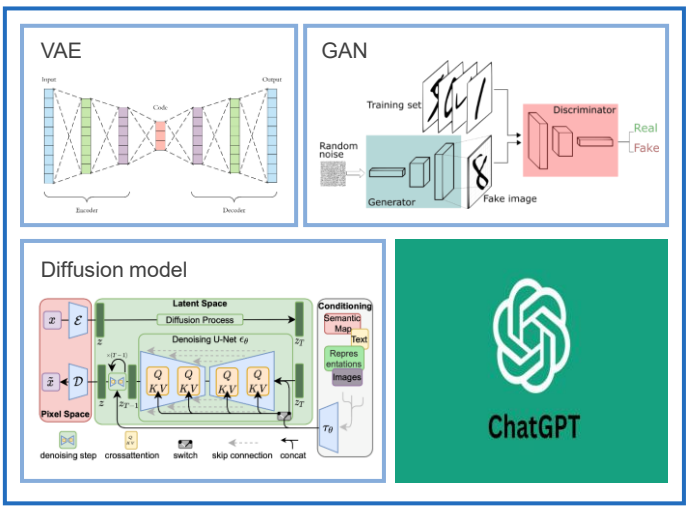
DL models in this course

FUNDAMENTAL

~~DISCRIMINATIVE~~



GENERATIVE



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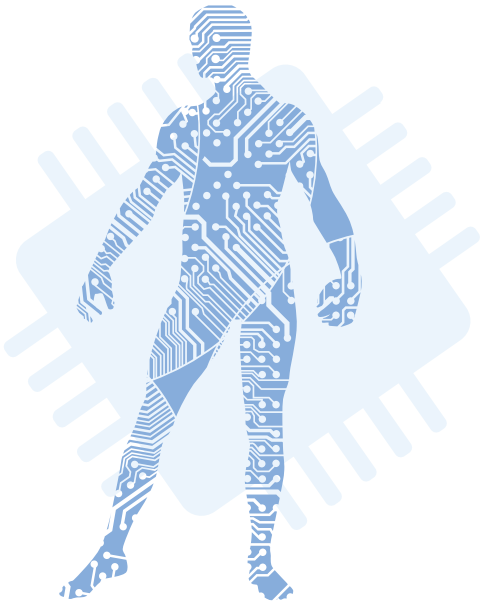
OUTLINE



01 Forward Pass
Learn MLP's components relating to the forward propagation

02 Backward Pass
Learn MLP's components relating to the backward propagation

03 Coding
Combine everything and implement a program with Keras

4

FWD Pass

Learn MLP's components relating to the forward propagation

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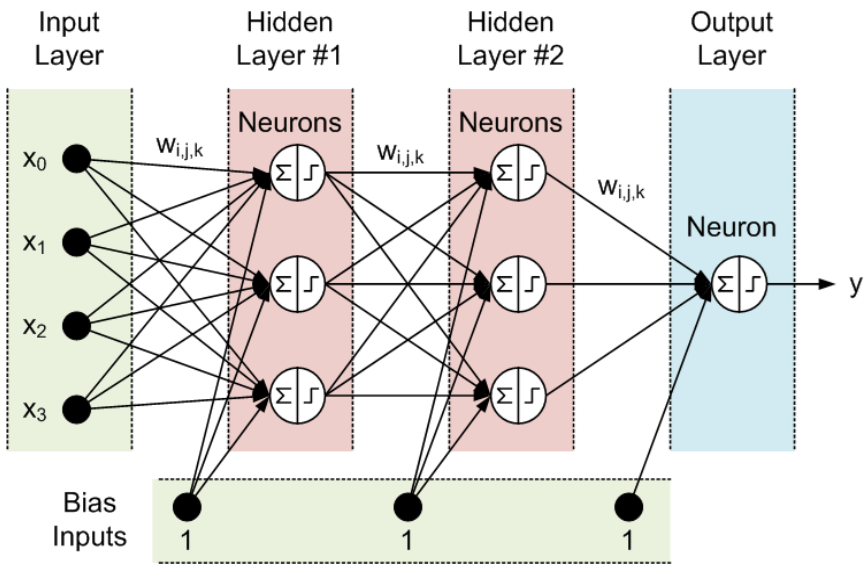
MULTI-LAYER PERCEPTRON

- **Layer**
  - Input layer
  - Output layer
  - Hidden layer
- **Neuron (node)**
  - Linear operation
  - Activation function
- **Trainable parameter**
  - Weight
  - Bias (in TensorFlow and PyTorch, there is one bias value per neuron)



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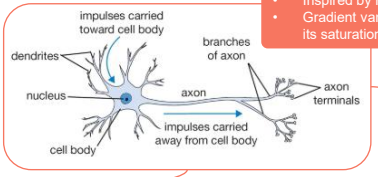
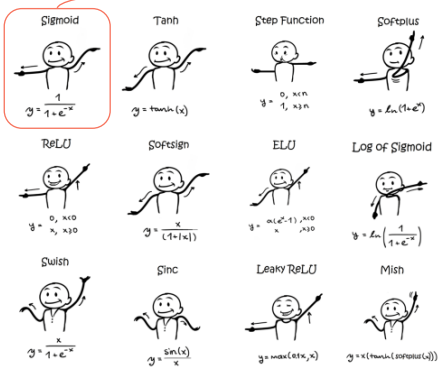
ACTIVATION FUNCTION

- Inspired by mother nature
- Gradient vanishing due to its saturation nature





Name	Plot	Equation	Derivative
Identity		$f(x) = x$ Equal to no activation function	$f'(x) = 1$
Binary step		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x \neq 0 \\ ? & \text{for } x = 0 \end{cases}$
Logistic (a.k.a Soft step)		$f(x) = \frac{1}{1+e^{-x}}$	$f'(x) = f(x)(1-f(x))$
TanH		$f(x) = \tanh(x) = \frac{2}{1+e^{-2x}} - 1$	$f'(x) = 1 - f(x)^2$
ArcTan		$f(x) = \tan^{-1}(x)$	$f'(x) = \frac{1}{x^2+1}$
Rectified Linear Unit (ReLU)		$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$
Parameteric Rectified Linear Unit (PReLU) [2]		$f(x) = \begin{cases} \alpha x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} \alpha & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$
Exponential Linear Unit (ELU) [3]		$f(x) = \begin{cases} \alpha(e^x - 1) & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} f(x) + \alpha & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$
SoftPlus		$f(x) = \log_e(1+e^x)$	$f'(x) = \frac{1}{1+e^{-x}}$

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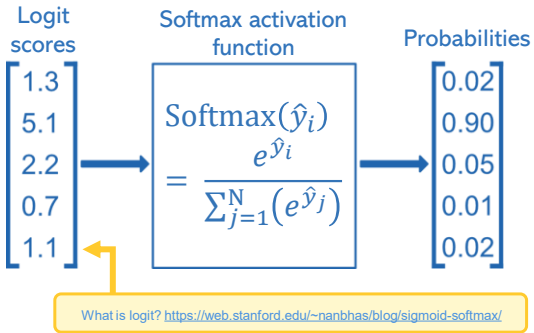
ACTIVATION FUNCTION

- Softmax activation function:
  - It is popular as an activation function for the output layer in a multi-class classification task (N output nodes). Use of softmax in a hidden layer is possible but less common.
  - Softmax function is used to normalize the outputs, converting them from weighted sum values into probabilities that sum to one. Each value in the output of the softmax function is interpreted as the probability of membership for each class.
  - Softmax can be thought of as a probabilistic or "softer" version of the argmax function as it is a smoother version of the winner-takes-all activation model in which the unit with the largest input has output +1 while all other units have output 0.
  - Softmax classifier predicts only one class at a time (multi-class not multi-output), so it should be used only with mutually exclusive classes.



19OCT2020: https://machinelearningmastery.com/softmax-activation-function-with-python/

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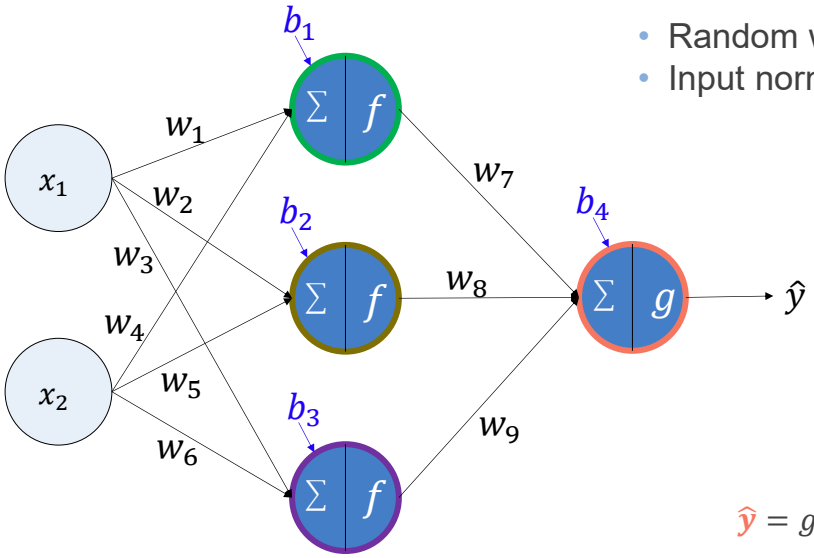
FORWARD PROPAGATION



- Random weight/bias initialization
- Input normalization/standardization

$$\boldsymbol{h_1} = f(w_1 x_1 + w_4\, x_2 + b_1)$$
$$\boldsymbol{h_2} = f(w_2 x_1 + w_5\, x_2 + b_2)$$
$$\boldsymbol{h_3} = f(w_3 x_1 + w_6 x_2 + b_3)$$
$$\hat{\boldsymbol{y}} = g(w_7 \boldsymbol{h_1} + w_8\, \boldsymbol{h_2} + w_9\, \boldsymbol{h_3} + b_4)$$

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CODE REVIEW

```
keras.layers.Dense(
    units,
    activation=None,
    use_bias=True,
    kernel_initializer="glorot_uniform",
    bias_initializer="zeros",
    kernel_regularizer=None,
    bias_regularizer=None,
    activity_regularizer=None,
    kernel_constraint=None,
    bias_constraint=None,
    lora_rank=None,
    lora_alpha=None,
    **kwargs
)
```

If set, help reduce the computation cost of fine-tuning large dense layers using the LoRA technique

`Dense` implements the operation: `output = activation(dot(input, kernel) + bias)` where `activation` is the element-wise activation function passed as the `activation` argument, `kernel` is a weights matrix created by the layer, and `bias` is a bias vector created by the layer (only applicable if `use_bias` is `True`). These are all attributes of `Dense`.

Note: If the input to the layer has a rank greater than 2, then `Dense` computes the dot product between the `inputs` and the `kernel` along the last axis of the `inputs` and axis 0 of the `kernel` (using `tf.tensordot`). For example, if input has dimensions `(batch_size, d0, d1)`, then we create a `kernel` with shape `(d1, units)`, and the `kernel` operates along axis 2 of the `input`, on every sub-tensor of shape `(1, 1, d1)` (there are `batch_size * d0` such sub-tensors). The output in this case will have shape `(batch_size, d0, units)`.

Synonyms:
- Dense layer (Keras, TensorFlow)
- Linear layer (PyTorch)
- Fully-connected layer

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BKWD Pass

Learn MLP's components relating to the backward propagation

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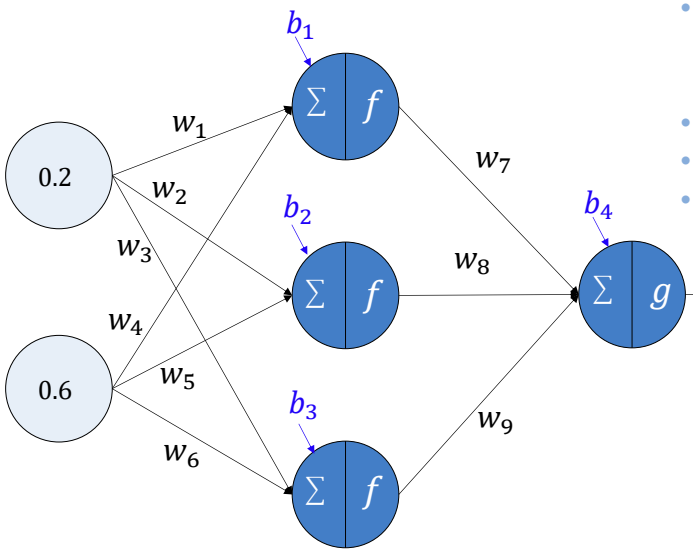
BACKWARD PROPAGATION (1986)



- Gradient Descent
  - Objective/Cost/Loss function
  - Partial derivation and chain rule
- Mini-batch gradient descent
- Epoch/Iteration
- Optimizer

$\hat{y} = 0.3$

Incorrect prediction

Training dataset

#	x_1	x_2	y
1	0.2	0.6	1.0
2	0.8	0.4	0.0
3	0.5	0.5	0.3

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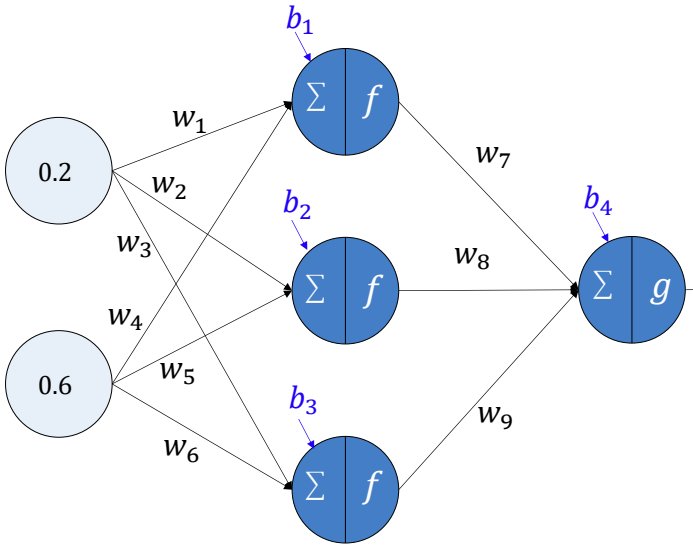
OBJECTIVE / COST / LOSS



- Objective function
- Cost function (often an average of the loss across the training set)
- Loss function (one training sample)

Predicted value

$\hat{y} = 0.3$

$y = 1.0$

Target value (ground truth, label)

Use a loss function to measure an error regarding one training sample

Training dataset

#	x_1	x_2	y
1	0.2	0.6	1.0
2	0.8	0.4	0.0
3	0.5	0.5	0.3

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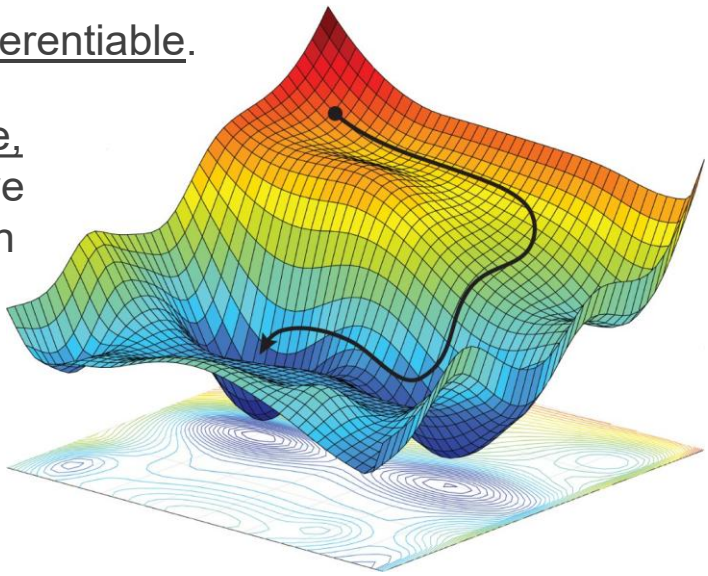
OBJECTIVE / COST / LOSS

- Loss function must be differentiable.
- The smaller the loss value, the better. Hence, negative loss values are better than zero loss values.
- Some loss functions have smoother surfaces than others.



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Some losses in Keras	Equation	Usage
Mean Absolute Error (MAE) loss or L1 loss	$\frac{1}{\text{output size}} \sum_{i=1}^{\text{output size}} \lvert y_i - \hat{y}_i \rvert$	• Regression when there are a lot of outliers in the training set
Mean Squared Error (MSE) loss or L2 loss	$\frac{1}{\text{output size}} \sum_{i=1}^{\text{output size}} (y_i - \hat{y}_i)^2$	• For most regression problems
Binary Cross-Entropy (BCE) loss	$-\sum_{i=1}^{\text{output size}} [y_i \cdot \log \hat{y}_i + (1 - y_i) \cdot \log(1 - \hat{y}_i)]$	• Binary classification • Multi-label binary classification
Categorical Cross-Entropy loss	$-\sum_{i=1}^{\text{output size}} (y_i \cdot \log \hat{y}_i)$	• Multi-class classification (expect labels to be provided in a one-hot format)
Sparse Categorical Cross-Entropy loss	• Same as the categorical cross-entropy but this sparse implementation saves memory when the label is sparse (the number of classes is very large). • Also it saves computational time: ○ When labels are mutually exclusive, in normal cross-entropy, there is a lot of log/dot/sum operations applied on $\hat{y}$ probabilities that will eventually be 0s. ○ Instead of doing log operations on all $\hat{y}$ and then dot with y, we can directly pick the probability from $\hat{y}$ at the index provided by y. Then apply log operation. No dot and sum operation is needed.	• Single-label multi-class classification (expect labels to be provided as integers)

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Cross-Entropy Loss

- **Cross-Entropy (CE) Loss:**
  - Measure the difference between two sets of two or more values that sum to 1.0.
  - Measure the difference between a correct probability distribution and a predicted distribution.

$$-\sum_{i=1}^{n\_classes} [y_i * \log(\hat{y}_i)]$$

#No.	y (ground truth)	$\hat{y}$ (prediction)	Correct?
1	(0.2, 0.5, 0.3)	(0.1, 0.3, 0.6)	No
2	(0.5, 0.4, 0.1)	(0.5, 0.3, 0.2)	Yes
3	(1, 0, 0)	(0.3, 0.6, 0.1)	No
4	(0, 0, 1)	(0.3, 0.3, 0.4)	Yes

$$\begin{aligned}\#1 &= -[0.2 * \log(0.1) + 0.5 * \log(0.3) + 0.3 * \log(0.6)] \\ &= -[(0.2 * -2.30) + (0.5 * -1.20) + (0.3 * -0.51)] \\ &= -[(-0.46) + (-0.60) + (-0.15)] \\ &= 1.21\end{aligned}$$

$$\begin{aligned}\#2 &= -[0.5 * \log(0.5) + 0.4 * \log(0.3) + 0.1 * \log(0.2)] \\ &= -[(0.5 * -0.69) + (0.4 * -1.20) + (0.1 * -1.61)] \\ &= -[(-0.35) + (-0.48) + (-0.16)] \\ &= 0.99\end{aligned}$$

$$\begin{aligned}\#3 &= -[1 * \log(0.3) + 0 * \log(0.6) + 0 * \log(0.1)] \\ &= -[(1 * -1.20) + 0 + 0] \\ &= 1.20\end{aligned}$$

$$\begin{aligned}\#4 &= -[0 * \log(0.3) + 0 * \log(0.3) + 1 * \log(0.4)] \\ &= -[0 + 0 + (1 * -0.92)] \\ &= 0.92\end{aligned}$$

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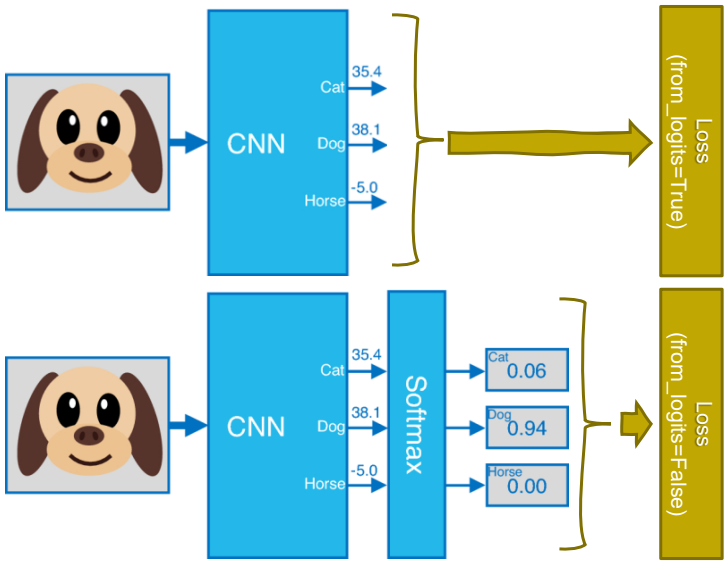
Cross-Entropy Loss



from_logits=True

- Internally apply Softmax in loss calculation
- Numerical stability

```
keras.losses.CategoricalCrossentropy(
    from_logits=False,
    label_smoothing=0.0,
    axis=-1,
    reduction="sum_over_batch_size",
    name="categorical_crossentropy",
    dtype=None,
)
```

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Cross-Entropy Loss

- **Cross-Entropy (CE) Loss for binary classification:**

$$-\sum_{i=1}^{n\_classes}[y_i * \log(\hat{y}_i)]$$

#No.	y (ground truth)	$\hat{y}$ (prediction)	Correct?
1	(1, 0)	(0.4, 0.6)	No
2	(0, 1)	(0.3, 0.7)	Yes
3	(0, 1)	(0.8, 0.2)	No

$$\#1 = -[1 * \log(0.4) + 0 * \log(0.6)] = -[(1 * -0.92) + 0] = 0.92$$

$$\#2 = -[0 * \log(0.3) + 1 * \log(0.7)] = -[0 + (-0.36)] = 0.36$$

$$\#3 = -[0 * \log(0.8) + 1 * \log(0.2)] = -[0 + (-1.61)] = 1.61$$

- **Binary Cross-Entropy (BCE) Loss:**
  - In the context of binary classification, knowing y and $\hat{y}$ of one class automatically implies the values of y and $\hat{y}$ of the other class. Hence, there is no need to encode as pairs of values.
  - The CE equation can be reduced to the following equation, when y and $\hat{y}$ refer to probabilities of the positive class. See that when $y = 1$ (positive class) the first term is added to our loss whereas when $y = 0$ (negative class) the second term is instead added.

$$(y)(-\log(\hat{y})) + (1 - y)(-\log(1 - \hat{y}))$$

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Why not MSE loss for classification?

Network 1

#No.	y (ground truth)	$\hat{y}$ (prediction)	Correct?		CE Loss	MSE Loss
1	(0, 0, 1)	(0.3, 0.3, 0.4)	Yes	⇒	0.92	0.54
2	(0, 1, 0)	(0.3, 0.4, 0.3)	Yes	⇒	0.92	0.54
3	(1, 0, 0)	(0.1, 0.2, 0.7)	No	⇒	2.30	1.34
				AVG	**1.38**	**0.81**

Network 2

#No.	y (ground truth)	$\hat{y}$ (prediction)	Correct?		CE Loss	MSE Loss
1	(0, 0, 1)	(0.1, 0.2, 0.7)	Yes	⇒	0.36	0.14
2	(0, 1, 0)	(0.1, 0.7, 0.2)	Yes	⇒	0.36	0.14
3	(1, 0, 0)	(0.3, 0.4, 0.3)	No	⇒	1.20	0.74
				AVG	**0.64**	**0.34**

- Both networks have the same classification error of 1/3=0.33, or equivalently a classification accuracy of 2/3 = 0.67. **BUT** the second network is better because it nails the first two training items and just barely misses the third training item.
- MSE is not a bad approach but compared to CE, MSE gives too much emphasis to the incorrect outputs while CE rewards/penalizes probabilities of correct classes only.

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OBJECTIVE / COST / LOSS

Regression losses

- MeanSquaredError class
- MeanAbsoluteError class
- MeanAbsolutePercentageError class
- MeanSquaredLogarithmicError class
- CosineSimilarity class
- Huber class
- LogCosh class
- Tversky class
- Dice class
- mean_squared_error function
- mean_absolute_error function
- mean_absolute_percentage_error function
- mean_squared_logarithmic_error function
- cosine_similarity function
- huber function
- log_cosh function
- tversky function
- dice function

Probabilistic losses

- BinaryCrossentropy class
- BinaryFocalCrossentropy class
- CategoricalCrossentropy class
- CategoricalFocalCrossentropy class
- SparseCategoricalCrossentropy class
- Poisson class
- CTC class
- KLDivergence class
- binary_crossentropy function
- categorical_crossentropy function
- sparse_categorical_crossentropy function
- poisson function
- ctc function
- kl_divergence function

Hinge losses for "maximum-margin" classification

- Hinge class
- SquaredHinge class
- CategoricalHinge class
- hinge function
- squared_hinge function
- categorical_hinge function

https://keras.io/api/losses/

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OBJECTIVE / COST / LOSS

`nn.L1Loss`	Creates a criterion that measures the mean absolute error (MAE) between each element in the input x and target y.
`nn.MSELoss`	Creates a criterion that measures the mean squared error (squared L2 norm) between each element in the input x and target y.
`nn.CrossEntropyLoss`	This criterion computes the cross entropy loss between input logits and target.
`nn.CTCLoss`	The Connectionist Temporal Classification loss.
`nn.NLLLoss`	The negative log likelihood loss.
`nn.PoissonNLLLoss`	Negative log likelihood loss with Poisson distribution of target.
`nn.GaussianNLLLoss`	Gaussian negative log likelihood loss.
`nn.KLDivLoss`	The Kullback-Leibler divergence loss.
`nn.BCELoss`	Creates a criterion that measures the Binary Cross Entropy between

https://docs.pytorch.org/docs/stable/nn.html#loss-functions

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Harmonic Loss to Replace CE Loss

- **Harmonic loss** has two desirable mathematical properties that enable faster convergence and improved interpretability:
  1) Scale invariance
  2) A finite convergence point, which can be interpreted as a class center

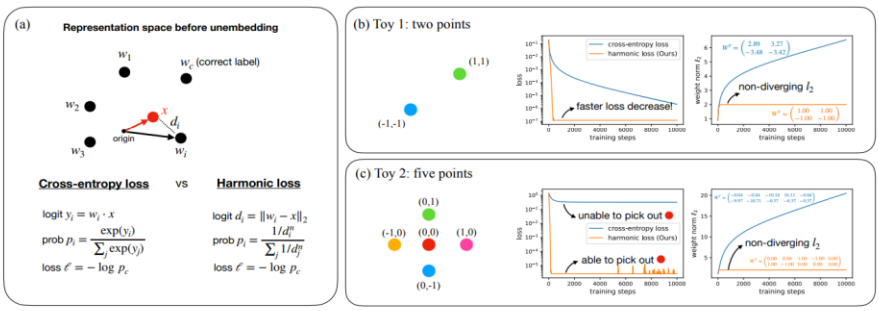

Figure 1. Cross-entropy loss versus harmonic loss (ours). (a) Definitions. Cross-entropy loss leverages the inner product as the similarity metric, whereas the harmonic loss uses Euclidean distance. (b) Toy case 1 with two points (classes). Both the harmonic loss and the l_2 weight norm converge faster for the harmonic loss. (c) Toy case 2 with five points (classes). Harmonic loss can pick out the red point in the middle. By contrast, the cross-entropy loss cannot, since the red point is not linearly separable from other points. The weight matrices are also more interpretable with harmonic loss than with cross-entropy loss.

MIT, "Harmonic Loss Trains Interpretable AI Models," arXiv 3FEB2025 https://arxiv.org/abs/2502.01628

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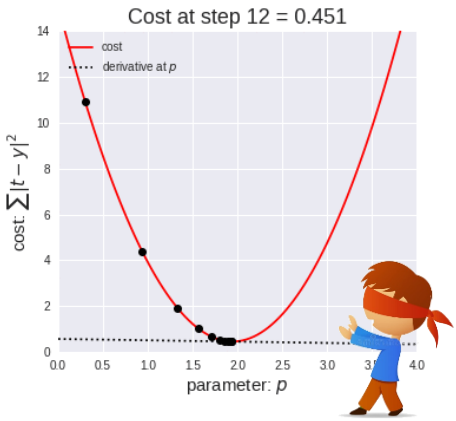
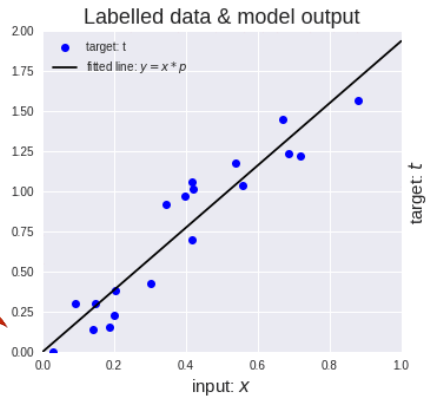
GRADIENT DESCENT

- **Gradient descent** is an optimization algorithm based on a convex function and tweaks its parameters iteratively to minimize a given function to its local minimum.





$$W_{new} = W_{old} - \alpha * \frac{\partial(Loss)}{\partial(W_{old})}$$

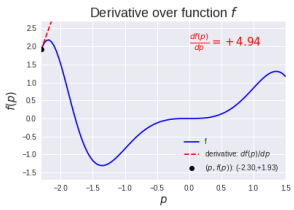


Image credit: https://medium.com/onfido-tech/machine-learning-101-be2e0a86c96a

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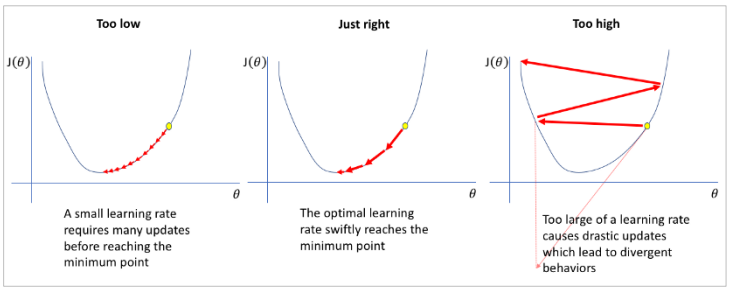
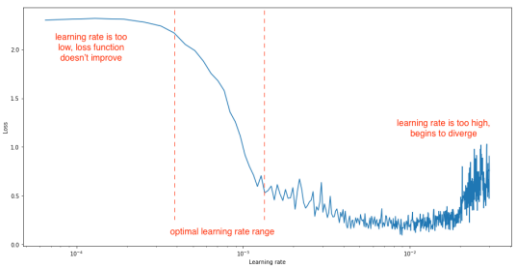
GRADIENT DESCENT

- How big/small the steps are gradient descent takes into the direction of the local minimum are determined by the learning rate, which figures out how fast or slow we will move towards the optimal weights.





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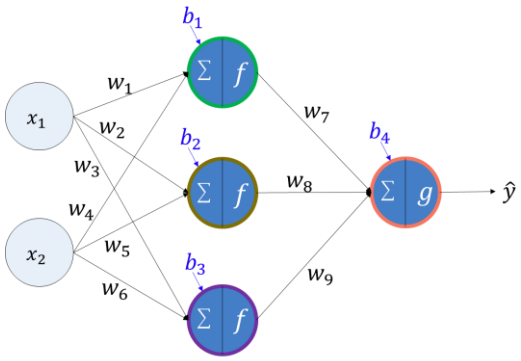
BACKWARD PROPAGATION (1986)

$$w_i \leftarrow w_i - \left(lr * \frac{\partial(\text{Loss})}{\partial w_i}\right) \quad , \quad b_i \leftarrow b_i - \left(lr * \frac{\partial(\text{Loss})}{\partial b_i}\right)$$



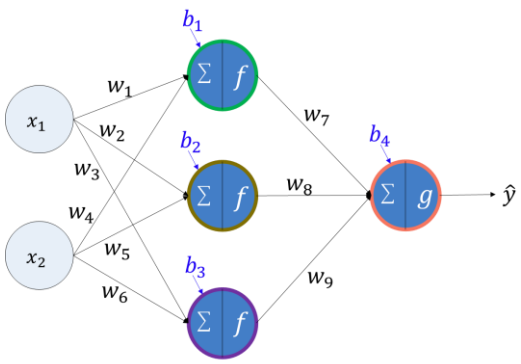
Example of updating w_7:

1. To update w_7, we have to compute
   $$w_7 = w_7 - \left(lr * \frac{\partial(\text{Loss})}{\partial w_7}\right)$$
2. In order to apply the chain rule, we first unfold the loss function until reaching w_7:
   $$\text{Loss} = (y - \hat{y})^2 \quad \texttt{\# use a squared error for simplification}$$
   $$\hat{y} = \boldsymbol{h_1} w_7 + \boldsymbol{h_2} w_8 + \boldsymbol{h_3} w_9 + b_4$$
3. Compute the gradient value by finding the partial derivation of loss with respect to w_7:
   $$\frac{\partial(\text{Loss})}{\partial w_7} = \frac{\partial(\text{Loss})}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_7} = (2 \cdot (y - \hat{y}) \cdot (-1)) \cdot (\boldsymbol{h_1})$$
4. Update w_7 in the opposite direction of the gradient

25

BACKWARD PROPAGATION (1986)

$$w_i \leftarrow w_i - \left(lr * \frac{\partial(\text{Loss})}{\partial w_i}\right) \quad , \quad b_i \leftarrow b_i - \left(lr * \frac{\partial(\text{Loss})}{\partial b_i}\right)$$



Example of updating w_1:

1. To update w_1, we have to compute
   $$w_1 = w_1 - \left(lr * \frac{\partial(\text{Loss})}{\partial w_1}\right)$$
2. In order to apply the chain rule, we first unfold the loss function until reaching w_1:
   $$\text{Loss} = (y - \hat{y})^2 \quad \texttt{\# use a squared error for simplification}$$
   $$\hat{y} = \boldsymbol{h_1} w_7 + \boldsymbol{h_2} w_8 + \boldsymbol{h_3} w_9 + b_4$$
   $$\boldsymbol{h_1} = x_1 w_1 + x_2 w_4 + b_1$$
3. Compute the gradient value by finding the partial derivation of loss with respect to w_1:
   $$\frac{\partial(\text{Loss})}{\partial w_1} = \frac{\partial(\text{Loss})}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h_1} \cdot \frac{\partial h_1}{\partial w_1} = (2 \cdot (y - \hat{y}) \cdot (-1)) \cdot (w_7) \cdot (x_1)$$
4. Update w_1 in the opposite direction of the gradient

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	Backpropagation	Gradient Descent
Definition	An algorithm for calculating the gradients of the cost function	Optimization algorithm used to find the weights that minimize the cost function
Requirements	Differentiation via the chain rule	•Gradient via Backpropagation •Learning rate
Process	Propagating the error backwards and calculating the gradient of the error function with respect to the weights	Descending down the cost function until the minimum point and find the corresponding weights

"To put it plainly, gradient descent is the process of using gradients to find the minimum value of the cost function, while backpropagation is calculating those gradients by moving in a backward direction in the neural network. Judging from this, it would be safe to say that gradient descent relies on backpropagation."

5APR2023: https://www.analyticsvidhya.com/blog/2023/01/gradient-descent-vs-backpropagation-whats-the-difference/

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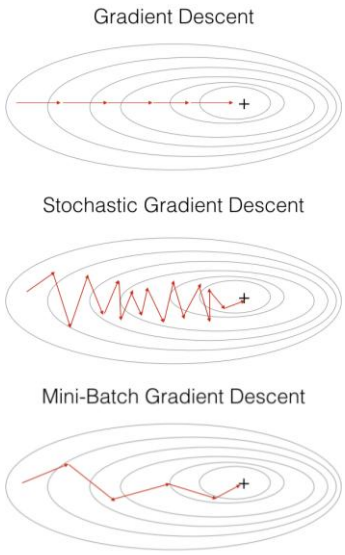
MINI-BATCH GRADIENT DESCENT

- Batch Gradient Descent:
  - Use the aggregated gradients from all training samples to update all trainable parameters
  - Slow calculation, large memory, and computationally expensive (due to large data)
- Stochastic Gradient Descent:
  - Use the gradient from one training sample to update all trainable parameters
  - Frequent updates and less memory, but noisy gradients, high variance, and computationally expensive (due to frequent updates)
- Mini-batch Gradient Descent:
  - Use the gradient from m training samples to update all trainable parameters, when m is the mini-batch size (`batch_size`)
  - More stable convergence, more efficient gradient calculation, and less memory but not guarantee good convergence



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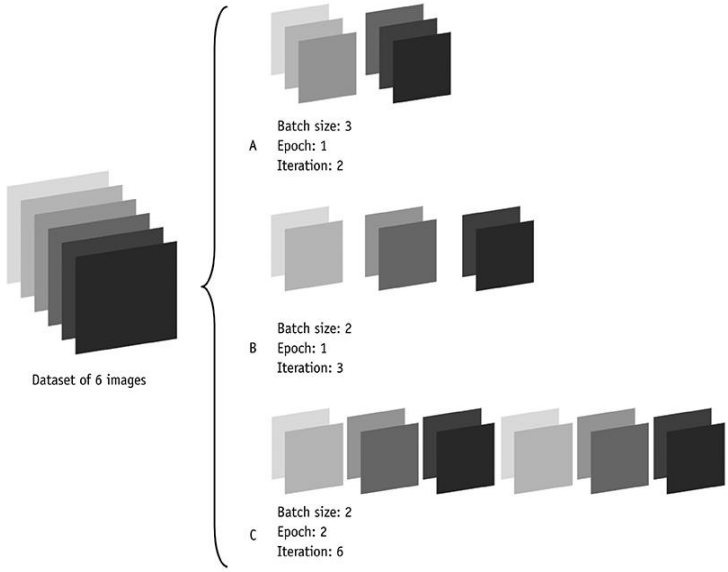
EPOCH / ITERATION

- One **epoch** is when an entire training dataset is passed forward and backward through the neural network only once.
- One **iteration** is when one (mini) batch of training samples is passed forward and backward through the neural network once.



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OPTIMIZER

- **Optimizers** are algorithms or methods used to minimize an error function (loss function) or to maximize the efficiency of production.
- Optimizers help get results faster.
- Gradient descent is an example of the optimizer.

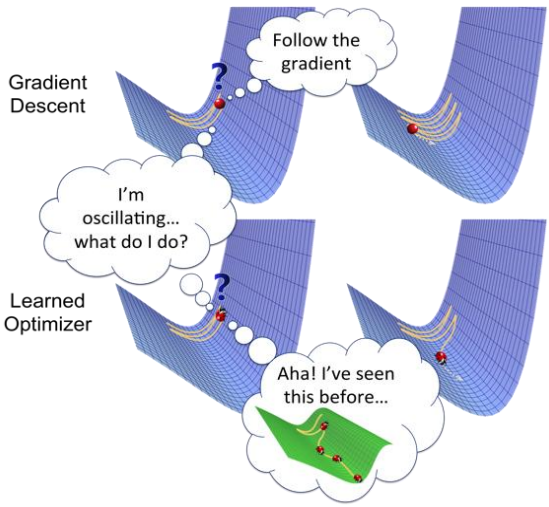

Image credit: (13JAN2019) https://towardsdatascience.com/optimizers-for-training-neural-network-59450d71caf6

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OPTIMIZER

Available optimizers

- SGD
- RMSprop
- Adam
- AdamW
- Adadelta
- Adagrad
- Adamax
- Adafactor
- Nadam
- Ftrl
- Lion
- Lamb
- Loss Scale Optimizer
- Muon

https://keras.io/api/optimizers/ , https://docs.pytorch.org/docs/stable/optim.html#algorithms

Adadelta	Implements Adadelta algorithm.
Adafactor	Implements Adafactor algorithm.
Adagrad	Implements Adagrad algorithm.
Adam	Implements Adam algorithm.
AdamW	Implements AdamW algorithm, where weight decay does not accumulate in the momentum nor variance.
SparseAdam	SparseAdam implements a masked version of the Adam algorithm suitable for sparse gradients.
Adamax	Implements Adamax algorithm (a variant of Adam based on infinity norm).
ASGD	Implements Averaged Stochastic Gradient Descent.
LBFGS	Implements L-BFGS algorithm.
NAdam	Implements NAdam algorithm.
RAdam	Implements RAdam algorithm.
RMSprop	Implements RMSprop algorithm.
Rprop	Implements the resilient backpropagation algorithm.
SGD	Implements stochastic gradient descent (optionally with momentum).

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FASTER OPTIMIZER

Available optimizers

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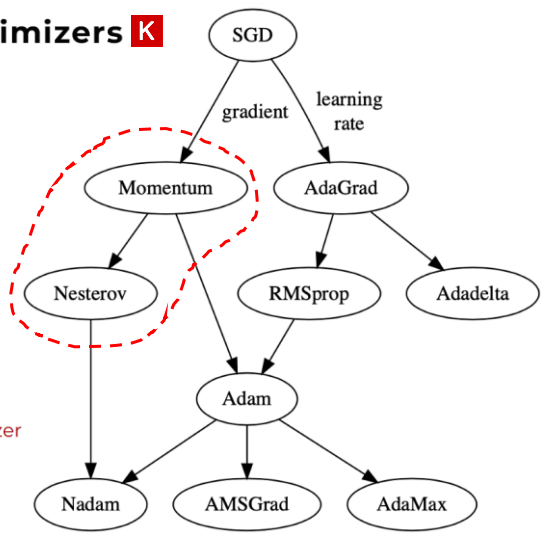

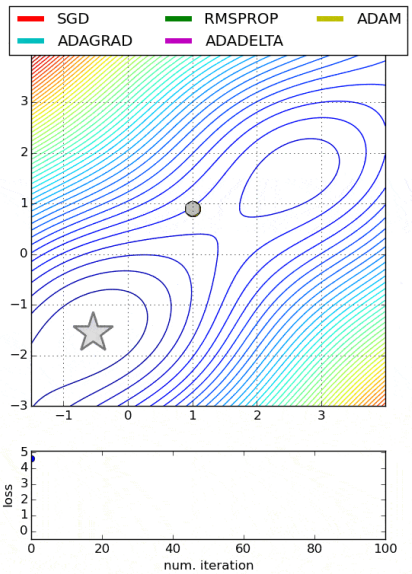

Conclusion diagram from https://www.kdnuggets.com/2019/06/gradient-descent-algorithms-cheat-sheet.html

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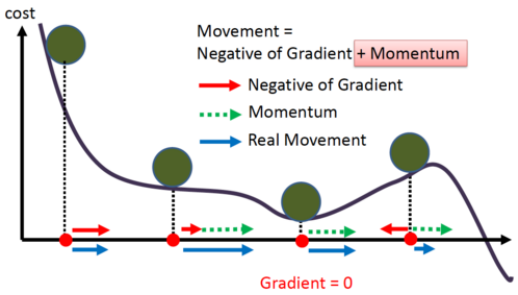
MOMENTUM (1964)

- **Vanilla Gradient Descent:**
  - It simply takes small, regular steps down the slope, so the algorithm will take much more time to reach the bottom.
  - It doesn't care much about what the earlier gradients were. If the local gradient is tiny, it goes very slowly.

- **Momentum in physics:**
  - An object with motion has some inertia which causes it to tend to move in the direction of motion.
  - Thus, if the optimization algorithm is moving in a general direction, the momentum causes it to 'resist' changes in direction.

- **Momentum optimization (1964):**
  - It cares much about what previous gradients were.
  - The update depends on **(the accumulated past) + (the gradient at that point)**.
  - It accelerates the convergence towards the relevant direction and reduces the fluctuation to the irrelevant direction.



33

MOMENTUM (1964)

- The first few updates may show no real advantage over vanilla Gradient Descent — since there are no previous gradients to utilize for the update.
- As the number of updates increases the momentum kickstarts and allows faster convergence.

- The momentum vector or the velocity
- The initial velocity is set to zero.

The local gradient is used for acceleration.

β is a hyperparameter called "momentum."
- Control how quickly the effect of past gradients decay
- Value between 0.0 (high friction) and 1.0 (no friction)
- A typical value is 0.9.

$$1.\ m_i \leftarrow \beta m_i - \left(lr * \frac{\partial(\text{Loss})}{\partial w_i}\right)$$

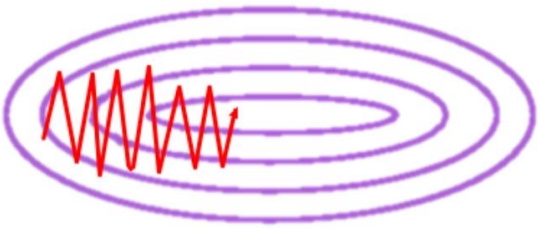
$$2.\ w_i \leftarrow w_i + m_i$$

34

MOMENTUM (1964)

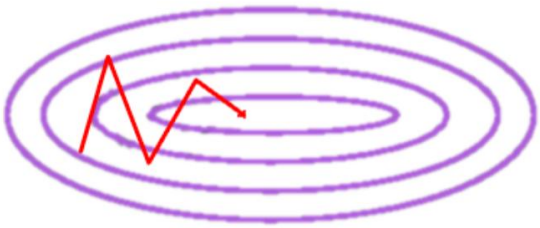
- To use momentum optimization, just use `keras.optimizers.SGD` and set its `momentum` hyperparameter

$$w_i \leftarrow w_i - \left(lr * \frac{\partial(\text{Loss})}{\partial w_i}\right)$$



$$\boldsymbol{m_i} \leftarrow \boldsymbol{\beta m_i} - \left(lr * \frac{\partial(\text{Loss})}{\partial w_i}\right)$$

$$w_i \leftarrow w_i + \boldsymbol{m_i}$$



(Left) Vanilla SGD, (right) SGD with momentum. Goodfellow et al. (2016)

35

NESTEROV MOMENTUM (1983)

- Problems of the vanilla momentum optimization:
  - The momentum problem near minima regions
  - Due to the momentum, the optimizer may overshoot a bit, then come back, overshoot again, and oscillate like this many times before stabilizing at the minimum.

- Nesterov Accelerated Gradient (NAG), also known as Nesterov momentum optimization, is a small variant to momentum optimization and is almost always faster than vanilla momentum optimization.
  - **Look before leap:** Measure the gradient of the cost function not at the local position w_i but slightly ahead in the direction of the momentum, at $w_i + \beta m$
  - This small tweak works because, in general, the momentum vector will be pointing in the right direction (toward the optimum), so it will be slightly more accurate to use the gradient measured a bit farther in that direction rather than the gradient at the original positions.

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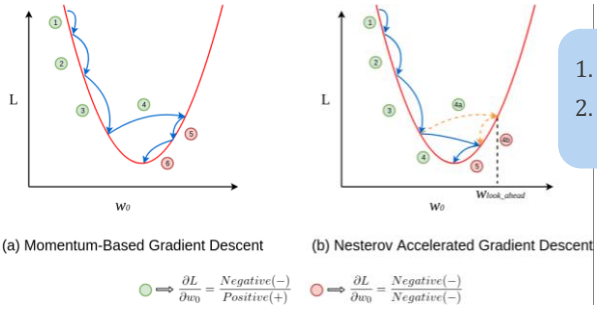
NESTEROV MOMENTUM (1983)

- When the momentum pushes the weights across a valley:
  - The gradient at the starting point w_i continues to push farther across the valley, while the gradient at the point $w_i + \beta m$ pushes back toward the bottom of the valley.
  - This helps reduce oscillations and thus NAG converges faster.

Momentum

1. $m_i \leftarrow \beta m_i - \left(lr * \frac{\partial(\text{Loss})}{\partial w_i}\right)$
2. $w_i \leftarrow w_i + m_i$



Nesterov Momentum

1. $m_i \leftarrow \beta m_i - \left(lr * \frac{\partial(\text{Loss})}{\partial(w_i + \beta m_i)}\right)$
2. $w_i \leftarrow w_i + m_i$

Look-ahead gradient

- To use NAG, simply set `nesterov=True` when creating the `keras.optimizers.SGD` optimizer

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FASTER OPTIMIZER

Available optimizers K

- SGD
- RMSprop
- Adam
- AdamW
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- Adamax
- Adafactor
- Nadam
- Ftrl
- Lion
- Lamb
- Loss Scale Optimizer
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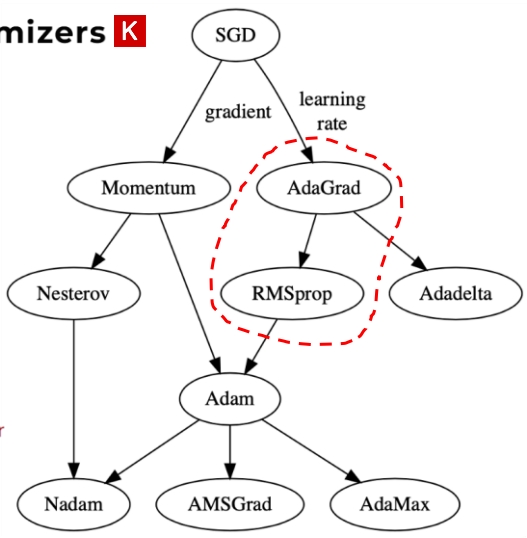

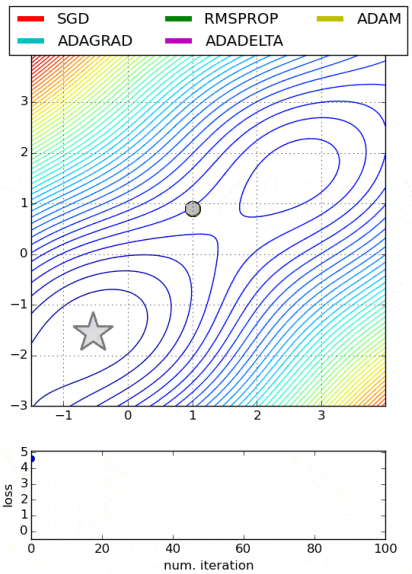

Conclusion diagram from https://www.kdnuggets.com/2019/06/gradient-descent-algorithms-cheat-sheet.html

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Adaptive Gradient (AdaGrad) (2011)

- Problem of the vanilla Gradient Descent:
  - Gradient descent starts by quickly going down the steepest slope, which does not point straight toward the global optimum, then it very slowly goes down to the bottom of the valley.
  - It would be nice if the algorithm could correct its direction earlier to point a bit more toward the global optimum.
- AdaGrad's Solution:
  - AdaGrad achieves the correction by scaling down the gradient vector along the steepest dimensions.
  - In short, AdaGrad decays the learning rate, but it does so faster for steep dimensions than for dimensions with gentler slopes. This is called an adaptive learning rate.

- s_i accumulates the squares of the loss function w.r.t. w_i.
- If the loss function is steep along the i^{th} dimension, then s_i will get larger and larger at each iteration.
- s is initialized by zero.

$$1.\ \boldsymbol{s_i} \leftarrow \boldsymbol{s_i} + \left(\frac{\partial(\text{Loss})}{\partial w_i}\right)^2$$

$$2.\ w_i \leftarrow w_i - \left(lr \cdot \frac{\partial(\text{Loss})}{\partial w_i} \cdot \frac{1}{\sqrt{\boldsymbol{s_i}+\varepsilon}}\right)$$

Scale down the gradient vector (a.k.a., learning rate decay)

A smooth term to avoid division by zero, typically set to 10^{-10}

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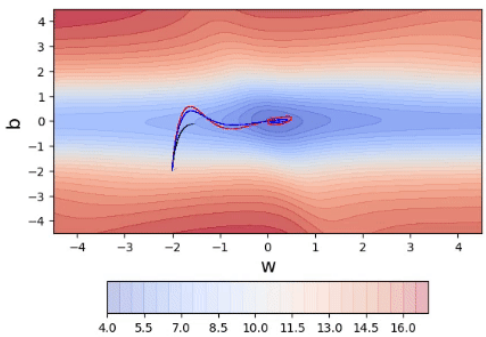
Adaptive Gradient (AdaGrad) (2011)

- AdaGrad's limitations:
  - It frequently performs well for simple quadratic problems, but it often stops too early when training neural networks. The learning rate gets scaled down so much that the algorithm ends up stopping entirely before reaching the global optimum.
  - It is recommended that we should not use AdaGrad to train deep neural networks (it may be efficient for simpler tasks such as Linear Regression, though).



Vanilla GD(black)
Momentum (red)
Nesterov (blue)
AdaGrad (green)

40

RMSProp (2012) by Hinton and Tieleman

- Problem of AdaGrad:
  - It runs the risk of slowing down a bit too fast and never converging to the global optimum.
- Solution of RMSProp:
  - Accumulate only the gradients from the most recent iterations (as opposed to all the gradients since the beginning of training) using exponential decay in the first step
  - To use RMSProp, please refer to `keras.optimizers.RMSprop`. Note that this implementation of RMSProp uses plain momentum, not Nesterov momentum.

β is a hyperparameter called "decay rate" or the `rho` hyperparameter in Keras.
- A typical value is 0.9 and often works well so we may not need to tune this at all.

AdaGrad

1. $s_i \leftarrow s_i + \left(\frac{\partial(\text{Loss})}{\partial w_i}\right)^2$
2. $w_i \leftarrow w_i - \left(lr \cdot \frac{\partial(\text{Loss})}{\partial w_i} \cdot \frac{1}{\sqrt{s_i}+\varepsilon}\right)$

RMSProp

1. $s_i \leftarrow \boldsymbol{\beta} s_i + (1-\boldsymbol{\beta})\left(\frac{\partial(\text{Loss})}{\partial w_i}\right)^2$
2. $w_i \leftarrow w_i - \left(lr \cdot \frac{\partial(\text{Loss})}{\partial w_i} \cdot \frac{1}{\sqrt{s_i}+\varepsilon}\right)$

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FASTER OPTIMIZER

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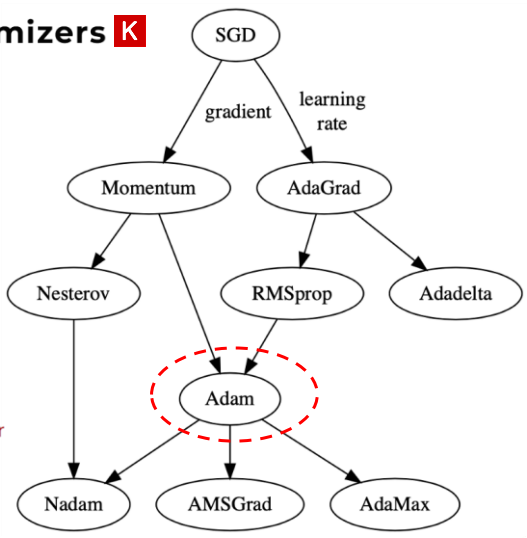

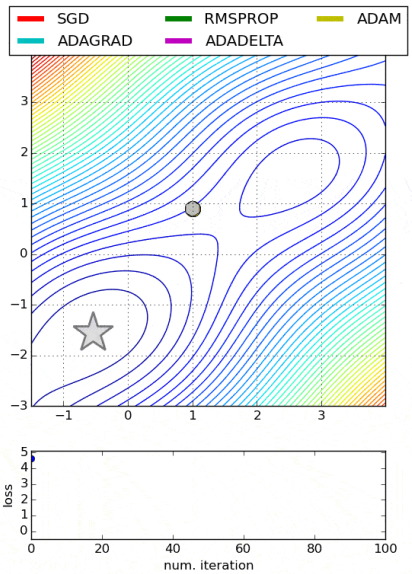

Conclusion diagram from https://www.kdnuggets.com/2019/06/gradient-descent-algorithms-cheat-sheet.html

42

Adam (arXiv 2014, ICLR 2015)

- Adam (Adaptive moment estimation) combines the ideas of momentum optimization and RMSProp.
  - Just like momentum optimization, Adam keeps track of an exponentially decaying average of past gradients.
  - Just like RMSProp, Adam keeps track of an exponentially decaying average of past squared gradients.
  - Since Adam is an adaptive learning rate algorithm (like AdaGrad and RMSProp), it requires less tuning of the learning rate hyperparameter lr. We can often use the default value of $lr = 0.001$, making Adam even easier to use than Gradient Descent.

- To use Adam, please refer to `keras.optimizers.Adam`:
  - The momentum decay hyperparameter `beta_1` is typically initialized to 0.9.
  - The scaling decay hyperparameter `beta_2` is often initialized to 0.999.
  - The smoothing term ε is usually initialized to a tiny number such as 10^{-7} (this is according to the default `keras.backend.epsilon()` value in Keras).

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Adam (arXiv 2014, ICLR 2015)

Momentum

1. $m_i \leftarrow \beta m_i - \left(lr * \frac{\partial(\text{Loss})}{\partial w_i}\right)$
2. $w_i \leftarrow w_i + m_i$

RMSProp

1. $s_i \leftarrow \beta s_i + (1-\beta)\left(\frac{\partial(Loss)}{\partial w_i}\right)^2$
2. $w_i \leftarrow w_i - \left(lr \cdot \frac{\partial(Loss)}{\partial w_i} \cdot \frac{1}{\sqrt{s_i}+\varepsilon}\right)$

An exponentially decaying average instead of an exponentially decaying sum

1. $m_i \leftarrow \boldsymbol{\beta_1} m_i + (1-\boldsymbol{\beta_1})\left(\frac{\partial(Loss)}{\partial w_i}\right)$
2. $s_i \leftarrow \boldsymbol{\beta_2} s_i + (1-\boldsymbol{\beta_2})\left(\frac{\partial(\text{Loss})}{\partial w_i}\right)^2$
3. $\hat{m}_i \leftarrow \frac{m_i}{1-\boldsymbol{\beta_1}^t}$
4. $\hat{s}_i \leftarrow \frac{s_i}{1-\boldsymbol{\beta_2}^t}$
5. $w_i \leftarrow w_i - \left(lr \cdot \frac{\hat{m}_i}{\sqrt{\hat{s}_i}+\boldsymbol{\varepsilon}}\right)$

- t represents the iteration number starting at 1.
- Since m and s are initialized at 0, they will be biased toward 0 at the beginning of training.
- These two steps will help boost m and s at the beginning of training.

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FASTER OPTIMIZER

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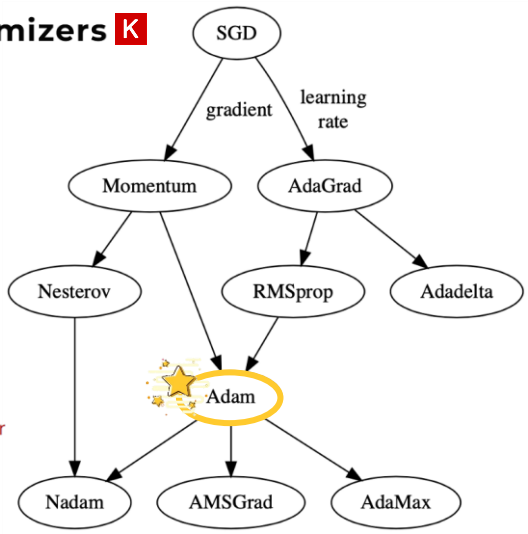

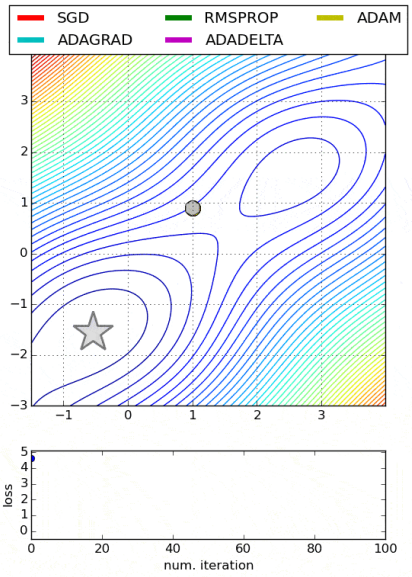

Conclusion diagram from https://www.kdnuggets.com/2019/06/gradient-descent-algorithms-cheat-sheet.html

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Adam with Weight Decay (arXiv 2017, ICLR 2019)

- **Adam (2014)** vs. **AdamW (2017):**
  - Both are adaptive optimizers widely used in deep learning.
  - Both are designed to manage momentum and adjust learning rates dynamically.
  - The difference is how they handle weight decay, which impacts their effectiveness in different scenarios.
- **Usage scenarios:** [21OCT2024: https://www.datacamp.com/tutorial/adamw-optimizer-in-pytorch]
  - Adam performs better in tasks where regularization is less critical, or when computational efficiency is prioritized over generalization.
    - For example Small neural networks (on datasets like MNIST or CIFAR-10), simple regression problems, early stage prototyping, less noisy data, short training cycles
  - AdamW excels in scenarios where overfitting is a concern and model size is substantial.
    - For example Large-scale transformers (like GPT), complex computer vision models (on datasets like ImageNet), multi-task learning, generative models (like GAN), reinforcement learning

I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," ICLR 2019, https://arxiv.org/abs/1711.05101

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Adam with Weight Decay (arXiv 2017, ICLR 2019)

- L2 Regularization vs. Weight Decay:
  - In SGD with momentum, L2 regularization is the same as weight decay.
    - L2 regularization adds $\boldsymbol{\lambda W_i}$ to the gradient-computation equation.
    - Weight decay adds $\boldsymbol{\lambda W_i}$ to the parameter-update equation.
  - **BUT**, in Adam, L2 regularization is not the same as weight decay.
    - L2 regularization leads to weights with large historic parameter and/or gradient amplitudes being regularized less than they would be when using weight decay.
    - It is suggested that, for Adam, we should use weight decay, not L2 regularization.

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Adam with Weight Decay (arXiv 2017, ICLR 2019)

Adam w/o regularization

0. $g_t \leftarrow \frac{\partial(Loss)}{\partial w_i}$
1. $m_i \leftarrow \boldsymbol{\beta_1} m_i + (1 - \boldsymbol{\beta_1})\, g_t$
2. $s_i \leftarrow \boldsymbol{\beta_2} s_i + (1 - \boldsymbol{\beta_2})(g_t)^2$
3. $\hat{m}_i \leftarrow \frac{m_i}{1-\boldsymbol{\beta_1}^t}$
4. $\hat{s}_i \leftarrow \frac{s_i}{1-\boldsymbol{\beta_2}^t}$
5. $w_i \leftarrow w_i - lr\left(\frac{\hat{m}_i}{\sqrt{\hat{s}_i}+\boldsymbol{\varepsilon}}\right)$

$\boldsymbol{\lambda}$ is the weight decay factor.

Adam with regularization

0. $g_t \leftarrow \frac{\partial(Loss + \boldsymbol{\lambda W_i})}{\partial w_i}$
1. $m_i \leftarrow \boldsymbol{\beta_1} m_i + (1 - \boldsymbol{\beta_1})\, g_t$
2. $s_i \leftarrow \boldsymbol{\beta_2} s_i + (1 - \boldsymbol{\beta_2})(g_t)^2$
3. $\hat{m}_i \leftarrow \frac{m_i}{1-\boldsymbol{\beta_1}^t}$
4. $\hat{s}_i \leftarrow \frac{s_i}{1-\boldsymbol{\beta_2}^t}$
5. $w_i \leftarrow w_i - lr\left(\frac{\hat{m}_i}{\sqrt{\hat{s}_i}+\boldsymbol{\varepsilon}}\right)$

- The standard and regularization gradients are coupled, mixing their effects during momentum calculation.
- This coupling can corrupt the direction and magnitude of updates.

AdamW with regularization

0. $g_t \leftarrow \frac{\partial(Loss)}{\partial w_i}$
1. $m_i \leftarrow \boldsymbol{\beta_1} m_i + (1 - \boldsymbol{\beta_1})\, g_t$
2. $s_i \leftarrow \boldsymbol{\beta_2} s_i + (1 - \boldsymbol{\beta_2})(g_t)^2$
3. $\hat{m}_i \leftarrow \frac{m_i}{1-\boldsymbol{\beta_1}^t}$
4. $\hat{s}_i \leftarrow \frac{s_i}{1-\boldsymbol{\beta_2}^t}$
5. $w_i \leftarrow w_i - lr\left(\frac{\hat{m}_i}{\sqrt{\hat{s}_i}+\boldsymbol{\varepsilon}} + \boldsymbol{\lambda W_i}\right)$

The regularized gradient is decoupled and applied after the smoothed standard gradient update to adjust the weights.

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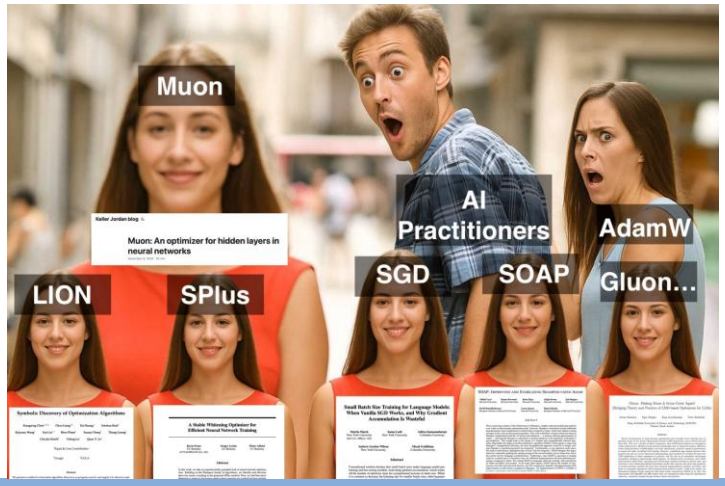
Muon optimizer

The second-order optimizer that surpasses AdamW in large-scale LLM training

- Since Adam (2014), new algorithms have come and gone without unseating Adam from its throne, with the exception of AdamW (2017).
- **Muon** is perhaps the leading example of how algorithms targeting specific NN training workloads are starting to make real headway against the formidable AdamW baseline.



12JUL2025: https://lukemerrick.com/posts/first_order_optimization.html

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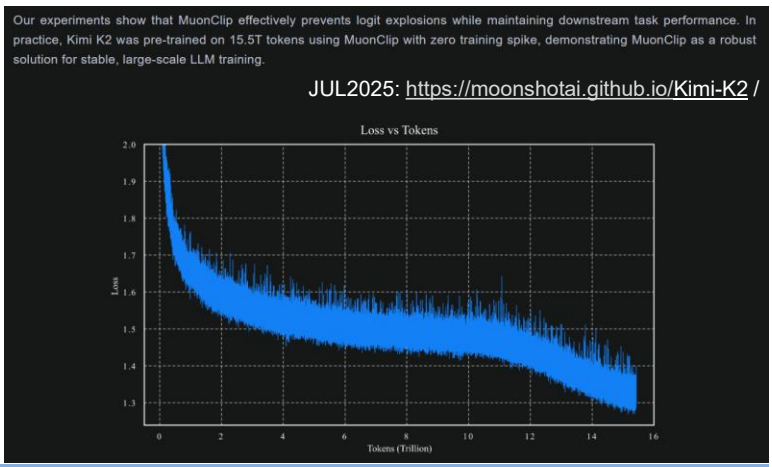
Muon optimizer

The second-order optimizer that surpasses AdamW in large-scale LLM training

- **Muon** made a splash by powering impressive gains on NanoGPT speed runs (OCT2024).
- **Muon** is having a big moment as the Kimi K2 model release (JUL2025) backs up the claims made in Moonshot's paper https://arxiv.org/abs/2502.16982 .



12JUL2025: https://lukemerrick.com/posts/first_order_optimization.html

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CODE REVIEW

```
Model.compile(
    optimizer="rmsprop",
    loss=None,
    loss_weights=None,
    metrics=None,
    weighted_metrics=None,
    run_eagerly=False,
    steps_per_execution=1,
    jit_compile="auto",
    auto_scale_loss=True,
)
```

```
keras.optimizers.SGD(
    learning_rate=0.01,
    momentum=0.0,
    nesterov=False,
    weight_decay=None,
    clipnorm=None,
    clipvalue=None,
    global_clipnorm=None,
    use_ema=False,
    ema_momentum=0.99,
    ema_overwrite_frequency=None,
    loss_scale_factor=None,
    gradient_accumulation_steps=None,
    name="SGD",
    **kwargs
)
```

```
keras.optimizers.Adam(
    learning_rate=0.001,
    beta_1=0.9,
    beta_2=0.999,
    epsilon=1e-07,
    amsgrad=False,
    weight_decay=None,
    clipnorm=None,
    clipvalue=None,
    global_clipnorm=None,
    use_ema=False,
    ema_momentum=0.99,
    ema_overwrite_frequency=None,
    loss_scale_factor=None,
    gradient_accumulation_steps=None,
    name="adam",
    **kwargs
)
```

```
keras.optimizers.RMSprop(
    learning_rate=0.001,
    rho=0.9,
    momentum=0.0,
    epsilon=1e-07,
    centered=False,
    weight_decay=None,
    clipnorm=None,
    clipvalue=None,
    global_clipnorm=None,
    use_ema=False,
    ema_momentum=0.99,
    ema_overwrite_frequency=None,
    loss_scale_factor=None,
    gradient_accumulation_steps=None,
    name="rmsprop",
    **kwargs
)
```

```
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=1e-3),
    loss=keras.losses.BinaryCrossentropy(),
    metrics=[
        keras.metrics.BinaryAccuracy(),
        keras.metrics.FalseNegatives(),
    ],
)
```

```
Model.fit(
    x=None,
    y=None,
    batch_size=None,
    epochs=1,
    verbose="auto",
    callbacks=None,
    validation_split=0.0,
    validation_data=None,
    shuffle=True,
    class_weight=None,
    sample_weight=None,
    initial_epoch=0,
    steps_per_epoch=None,
    validation_steps=None,
    validation_batch_size=None,
    validation_freq=1,
)
```

Assign different importance to each training sample/class when computing the loss, helping handle imbalanced or noisy data

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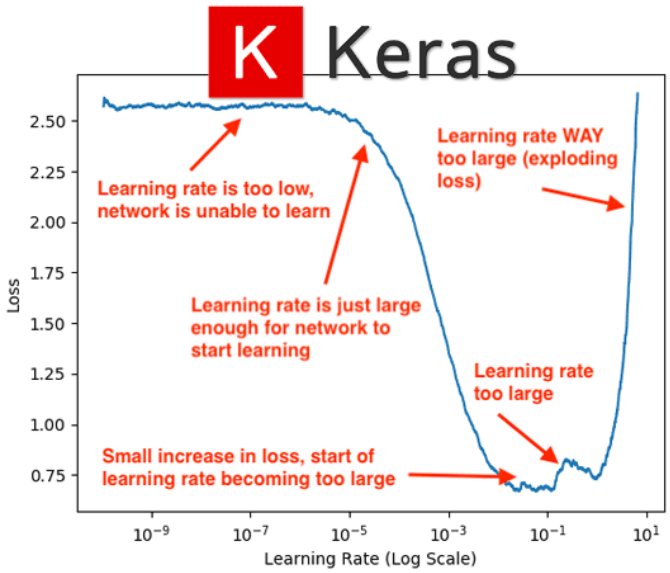
Learning Rate Tuning



5AUG2019:
https://pyimagesearch.com/2019/08/05/keras-learning-rate-finder/

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Learning Rate (LR) Scheduler

- LearningRateScheduler changes the learning rate as training progresses.
  - Early in training, the scheduler might keep the learning rate high to allow faster progress.
  - As the model improves, the scheduler reduces the learning rate to fine-tune the performance and prevent overshooting.

```
keras.callbacks.LearningRateScheduler(schedule, verbose=0)
```

```
>>> # This function keeps the initial learning rate for the first ten epochs
>>> # and decreases it exponentially after that.
>>> def scheduler(epoch, lr):
...     if epoch < 10:
...         return lr
...     else:
...         return lr * ops.exp(-0.1)
...
>>> model = keras.models.Sequential([keras.layers.Dense(10)])
>>> model.compile(keras.optimizers.SGD(), loss='mse')
>>> round(model.optimizer.learning_rate, 5)
0.01
```

```
>>> callback = keras.callbacks.LearningRateScheduler(scheduler)
>>> history = model.fit(np.arange(100).reshape(5, 20), np.zeros(5),
...                     epochs=15, callbacks=[callback], verbose=0)
>>> round(model.optimizer.learning_rate, 5)
0.00607
```

https://keras.io/api/callbacks/learning_rate_scheduler/

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Learning Rate (LR) Scheduler

- ReduceLROnPlateau reduces learning rate when a metric has stopped improving.
  - Models often benefit from reducing the learning rate by a factor of 2-10 once learning stagnates.
  - This callback monitors a quantity and if no improvement is seen for a 'patience' number of epochs, the learning rate is reduced.

```
keras.callbacks.ReduceLROnPlateau(
    monitor="val_loss",
    factor=0.1,
    patience=10,
    verbose=0,
    mode="auto",
    min_delta=0.0001,
    cooldown=0,
    min_lr=0.0,
    **kwargs
)
```

```
reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.2,
                              patience=5, min_lr=0.001)
model.fit(x_train, y_train, callbacks=[reduce_lr])
```

https://keras.io/api/callbacks/reduce_lr_on_plateau/

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Adam vs. LR Scheduler

- An adaptive-LR optimizer (like Adam) helps with per-parameter learning rates.
  - It updates each trainable parameter with an individual learning rate. This means that every parameter in the network has a specific learning rate associated with it.
  - **BUT** the single learning rate for each parameter is computed using lambda (the initial learning rate) as an upper limit. This means that every single learning rate can vary from 0 (no update) to lambda (maximum update).
- LR Scheduler refers to a global learning rate multiplier, giving us additional control over the overall training dynamics, improving both performance and stability.

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Adam + LR Scheduler

- Further finetuning:
  - Adam adjusts the learning rate individually for each parameter based on the gradient, but the overall learning rate still affects how much the model updates in each step.
  - An LR scheduler allows us to adjust the global learning rate over time, providing better control, especially during fine-tuning phases when smaller learning rates are crucial.
- Prevent overshooting:
  - While Adam adjusts the learning rates per parameter, there can still be cases where the learning rate is too high for some stages of training.
  - A scheduler can gradually decrease the learning rate as the model converges, helping avoid overshooting the minimum and settling into a better solution.
- Improve generalization:
  - A common technique, such as learning rate warmup or decay, can help models generalize better.
  - By slowly reducing the learning rate over time, we reduce the chance of overfitting and help the model find a more optimal, generalized solution.
- Handling plateaus:
  - Sometimes, training plateaus (stops improving) even with Adam.
  - An LR scheduler can lower the learning rate at key moments to help the model "escape" these plateaus and keep improving.

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Coding

Combine everything and implement a program with Keras

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EX 1: Your First MLP

- Use Keras (PyTorch backend) to create an MLP model based on a dummy dataset data
- **Seeding values:**
  - During training, set them to fixed values in order not to get variable results due to the random weight initialization
  - To conclude the proposed model's performance, you need to train the same model with the same set of selected hyperparameters repeatedly (at least 3-5 times, the more the better) using random seeds to average the obtained performances.

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EX 2: Hand-designed MLPs

- Four separated problems:
  - Regression with 4 numeric inputs in order to produce 3 numeric outputs
  - Binary classification with 4 numeric inputs in order to produce the yes-or-no prediction
  - Single-label multi-class classification with 4 numeric inputs in order to produce likelihoods regarding 6 output classes
  - Multi-label multi-class classification with 4 numeric inputs in order to produce likelihoods regarding 6 output classes
- Hyperparameter checklist:
  - The number of layers: input, hidden, output
  - The number of neurons in each layer
  - ~~Weight initialization~~
  - Input normalization
  - Activation function in each layer
  - ~~Regularization~~
  - Loss function
  - Optimizer
  - Initial learning rate
  - ~~Learning rate scheduler~~
  - (Mini-) Batch size
  - The number of epochs

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